GCI World

In-class Competition: Strategy and Tactics

松尾・岩澤研究室
MATSUO-IWASAWA LAB UTOKYO

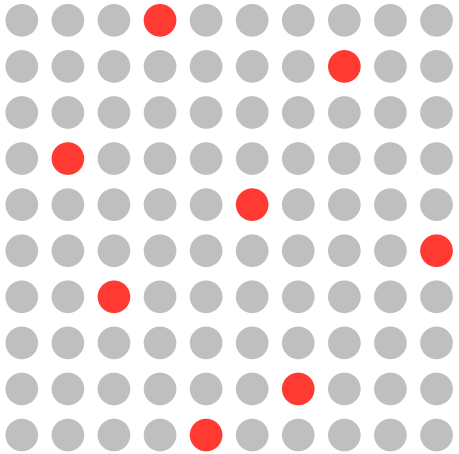
Photography, video recording and disclosure to third parties without permissions are strictly prohibited.

©MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

What Exactly Is Being Scored?

Only 8 in 100 default, so accuracy is fooled. **AUC scores the ranking: submit probabilities.**

1 Accuracy is fooled

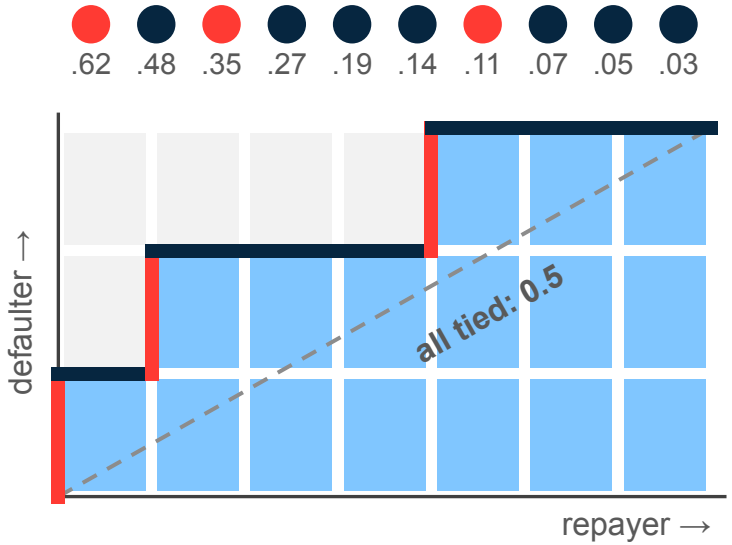


8 in 100 default

Saying “nobody defaults” is **91.9% accurate**

2 AUC scores the ranking

Sorted by predicted probability, high → low



Each cell = one (defaulter, repayer) pair

3 Blue share = AUC

Blue cell: that defaulter is ranked above that repayer

16 / 21 = 0.76

Submit probabilities, not 0/1

`predict_proba(X)[:, 1]`	**0.730**
`predict(X)`	**0.500**

predict() says 0 for everyone: all tied, just like “nobody defaults”

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

Treat Every Idea as a Hypothesis

Right or wrong, **turn each result into your next hypothesis.**

4 → 1 Why did it happen? → What does that say about the data or the model? → What does that predict next?

Example log LightGBM on the competition data, the same 5 folds, each try on top of the last kept one

	Try 1 hypothesis wrong	Try 2 hypothesis right	Try 3 hypothesis wrong
1 Hypothesize	It can't tell that 365243 means "not employed"	It never sees education or job type, which signal risk	Filling gaps with 0 hides who has no bureau record
2 Experiment	Replace 365243 with NaN	Add the 6 text columns	Flag it before the fill
3 Analyze	**1 of 5 folds better → drop** average -0.0005	**5 of 5 folds better → keep** average +0.0032	**3 of 5 folds better → drop** average +0.0002
4 Revise	**A tree splits 365243 off by itself** **→ What can't it see at all?**	**The model had never seen them** **→ What else isn't it seeing?**	**It already knew: they all lack EXT_SOURCE_3** **→ Is my next idea new to the model?**

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

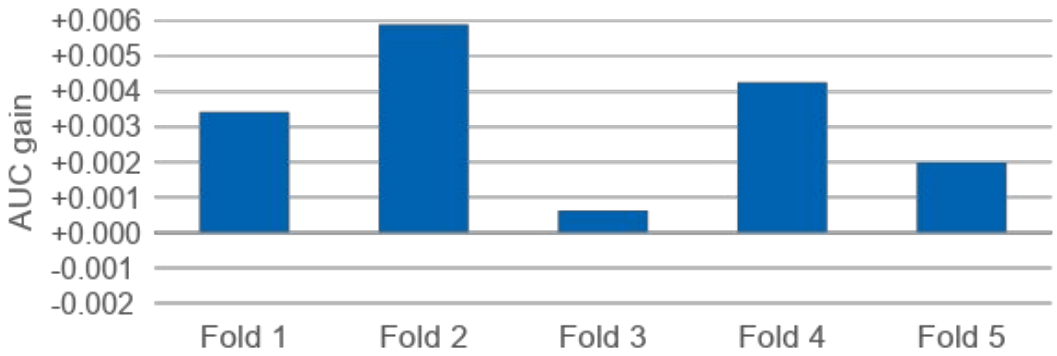
Good Improvement vs Bad Improvement

A real gain wins fold after fold, **not just on average.**

✓ Good improvement

+ 6 text columns (education, job, …)

LightGBM. Each bar: AUC with the change − without it, on one fold



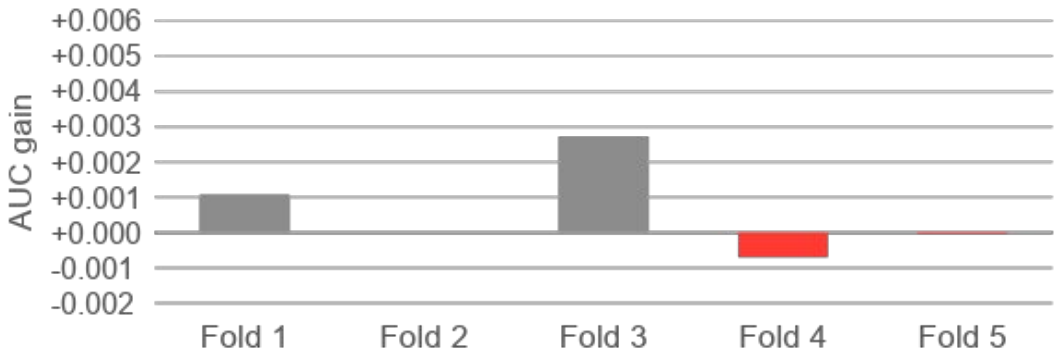
Better on all 5 folds

average gain +0.0032

✗ Bad improvement

+ AMT_CREDIT / AMT_GOODS_PRICE

LightGBM. Each bar: AUC with the change − without it, on one fold



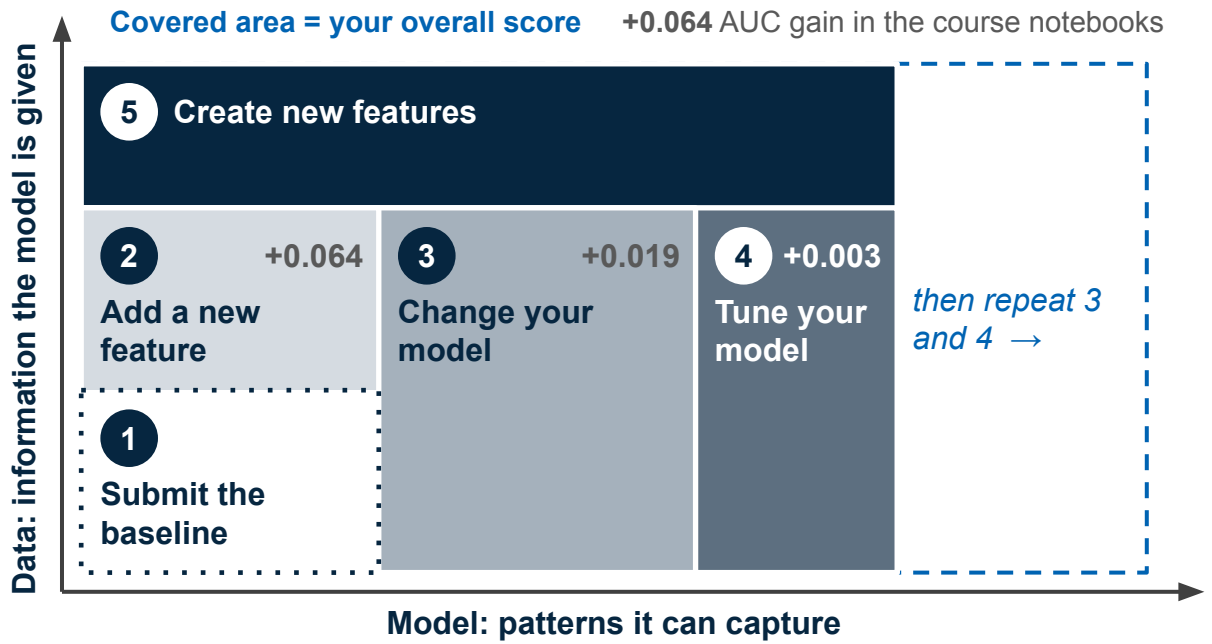
Up on average, not fold by fold

average gain +0.0006

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

Why This Order? Two Axioms: Data and Model

Every step grows your score along one axis: **data first, then the model, then repeat.**



Axiom 1 · Data

A model can't use information it isn't given.

↑ **Steps 2 and 5 grow the area upward**

Axiom 2 · Model

A model captures only the patterns its capacity allows.

→ **Steps 3 and 4 grow it to the right**

Why data first? More capacity can't find what the data doesn't hold, and the best settings depend on the data you give it.

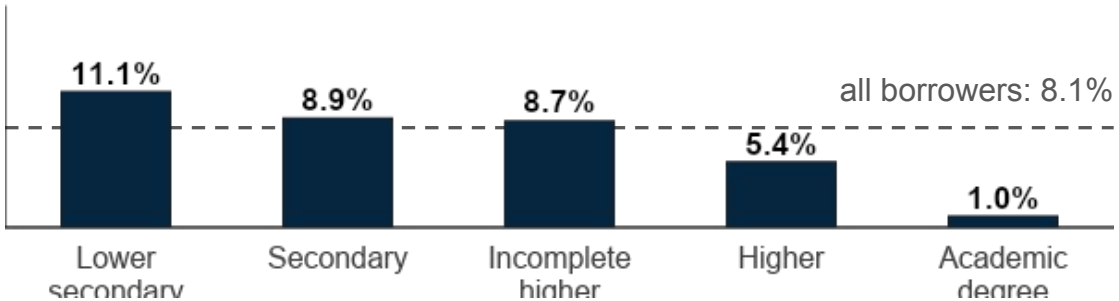
©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

What Does My Model Actually See?

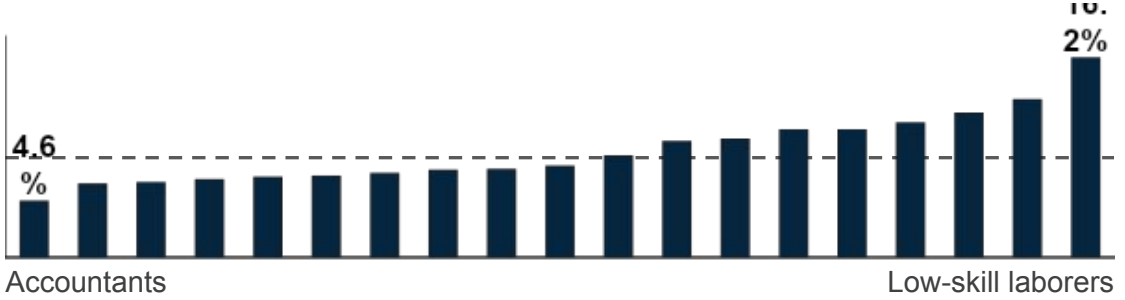
The notebook drops text columns that matter: **turn them into 0/1 columns.**

Do the 6 dropped text columns matter? Default rate by group

By education



By occupation (19 types, sorted)



The model takes numbers only, so the notebook drops text. One-hot encoding fixes that:

Text column: dropped

NAME_EDUCATION_TYPE
Higher education
Lower secondary
Incomplete higher

One 0/1 column per category: kept

Higher education	Lower secondary	Incomplete higher
1	0	0
0	1	0
0	0	1

```
train = pd.get_dummies(train, dtype=int)
```

In a new cell after Section 5; the same for test. It encodes all 6 text columns at once.

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

Why These Four Features?

Two habits find features: **think like a lender, and look for what looks odd.**

1 What would a lender check?

In the lender's chair: will this loan be paid back?

THINK	The same payment is easy on a high income and heavy on a low one.
ASK	**Can they afford each payment?**
BUILD	`ANNUITY_INCOME_RATIO` = annuity ÷ income
THINK	Two loans of the same size can be repaid over very different lengths of time.
ASK	**How many payments until it's repaid?**
BUILD	`CREDIT_TERM` = credit ÷ annuity

2 What looks odd in the data?

Plot each column and ask what the oddity does to a model.

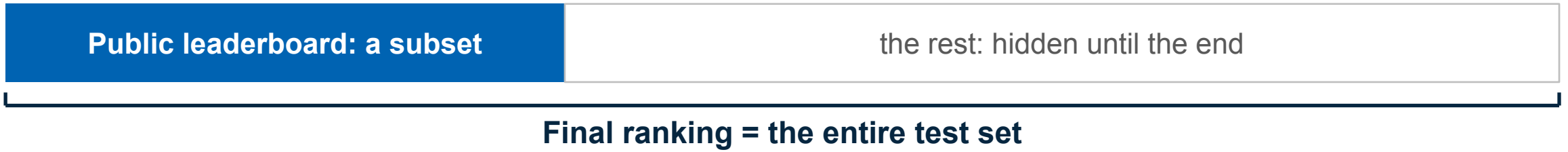
NOTICE	The outside scores predict best, but are often missing (EXT_SOURCE_1 for 70% of borrowers).
ASK	**Can three patchy scores become one?**
BUILD	`EXT_MEAN` = mean of the three scores
NOTICE	Income is extremely skewed: the largest is 92 times the median.
ASK	**Do a few huge incomes stretch the scale?**
BUILD	`LOG_AMT_INCOME_TOTAL` = log(1 + income)

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

What Is the Public Leaderboard Telling You?

The public score is a small, noisy sample. **Your last file is your final one.**

test.csv



Oct 2
start

Nov 20, 01:00 UTC
your LAST file counts

Dec 17 (Session 14)
ranking announced

Pick your final model by CV, and re-upload your best file at the end.

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

Rules of Thumb

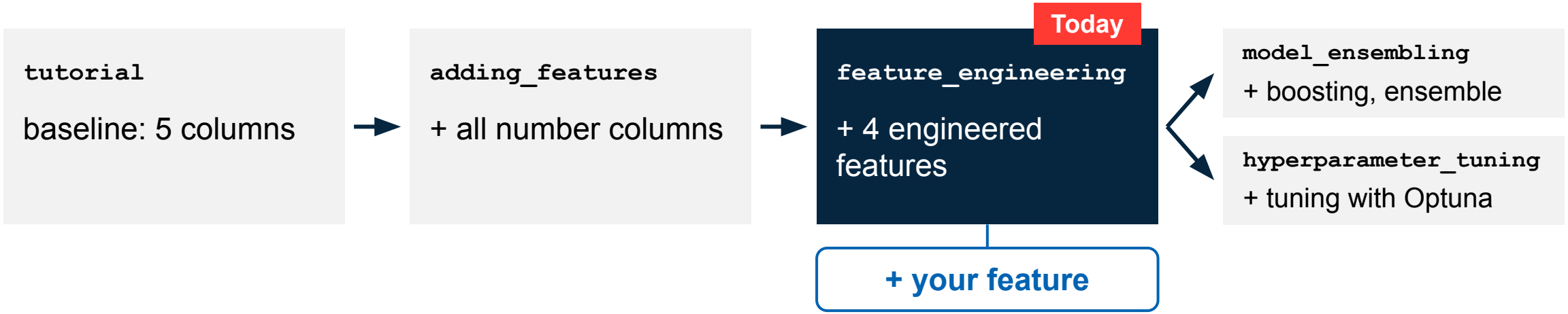
Small, honest, fast experiments beat big guesses.

✔ Do	✕ Don’t
Submit a simple baseline early	Wait for the perfect model
Look at what your model sees	Feed in columns you never checked
Change one thing, write it down	Change five things at once
Iterate fast on a sample	Wait an hour to test one idea
Trust CV on fixed folds	Chase the public leaderboard

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

The Notebooks Build on Each Other

Each notebook = the previous one + one change. **Today you add the next one.**



Then carry it forward: both later notebooks share the same `add_effective_features()`

©MATSUO–IWASAWA LAB, THE UNIVERSITY OF TOKYO

松尾・岩澤研究室

MATSUO-IWASAWA LAB UTOKYO

©MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO